ESERCIZIARIO QUANTUM

$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $\qquad$ $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Gate	Action on Computational Basis	Matrix Representation
Identity	$I\|0\rangle = \|0\rangle$ $I\|1\rangle = \|1\rangle$	$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
Pauli X	$X\|0\rangle = \|1\rangle$ $X\|1\rangle = \|0\rangle$	$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
Pauli Y	$Y\|0\rangle = i\|1\rangle$ $Y\|1\rangle = -i\|0\rangle$	$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$
Pauli Z	$Z\|0\rangle = \|0\rangle$ $Z\|1\rangle = -\|1\rangle$	$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
Phase S	$S\|0\rangle = \|0\rangle$ $S\|1\rangle = i\|1\rangle$	$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$
T	$T\|0\rangle = \|0\rangle$ $T\|1\rangle = e^{i\pi/4}\|1\rangle$	$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$
Hadamard H	$H\|0\rangle = \frac{1}{\sqrt{2}}(\|0\rangle + \|1\rangle)$ $H\|1\rangle = \frac{1}{\sqrt{2}}(\|0\rangle - \|1\rangle)$	$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

PROBABILITÀ

Esempio: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

$Pr(|0\rangle) = |\alpha|^2 \qquad Pr(|1\rangle) = |\beta|^2$

ESEMPIO CON 2 QBIT: $|\psi\rangle = \alpha|00\rangle + \beta|10\rangle$

$Pr(|00\rangle) = |\alpha|^2 \qquad Pr(|10\rangle) = |\beta|^2$

GATE CONTROLLATI

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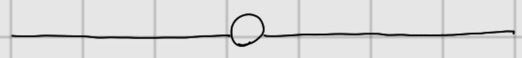
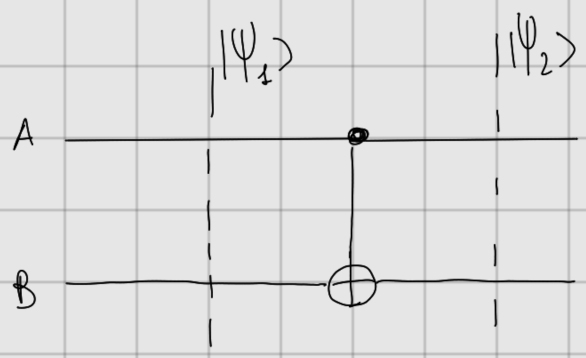
PALLINO PIENO $\longrightarrow$ TRIGGERA VALORE "1"



PALLINO VUOTO $\longrightarrow$ TRIGGERA VALORE "0"

ESEMPIO :



qbit di controllo

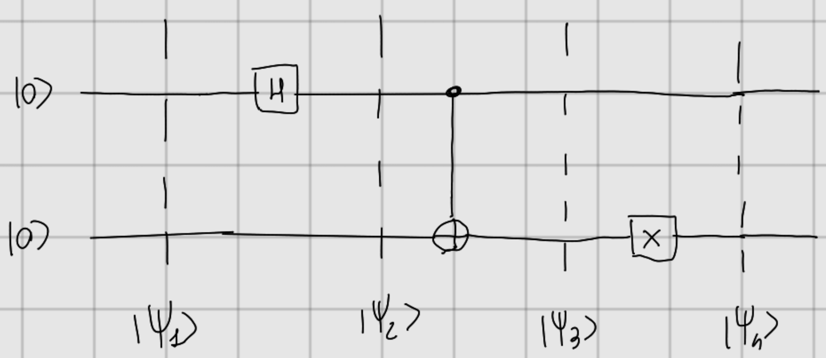
Se $|\psi_1\rangle = |00\rangle \longrightarrow |\psi_2\rangle = |00\rangle$

poiché qbit controllo $= \emptyset$

Se $|\psi_1\rangle = |10\rangle \longrightarrow |\psi_2\rangle = |11\rangle$

poiché qbit controllo $= 1$

ESEMPIO ESERCIZIO



$|\psi_1\rangle = |0\rangle|0\rangle$

$|\psi_2\rangle = H|0\rangle \otimes I|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle = \frac{|00\rangle + |10\rangle}{\sqrt{2}}$

$$|\psi_3\rangle = \frac{CNOT\,|00\rangle + CNOT\,|10\rangle}{\sqrt{2}} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

$$|\psi_4\rangle = \frac{I|0\rangle X|0\rangle + I|1\rangle X|1\rangle}{\sqrt{2}} = \frac{|0\rangle|1\rangle + |1\rangle|0\rangle}{\sqrt{2}}$$

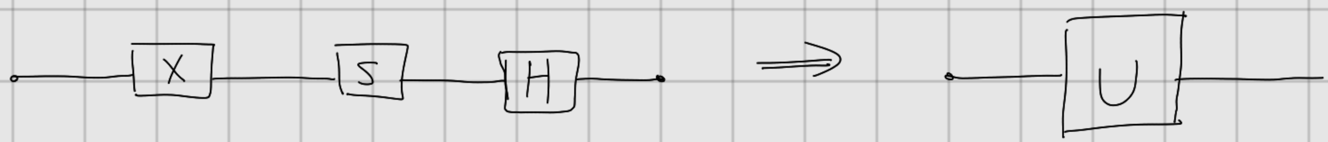
$$|\psi_4\rangle = \frac{1}{\sqrt{2}}|01\rangle + \frac{1}{\sqrt{2}}|10\rangle$$

$$Pr(|01\rangle) = \left|\frac{1}{\sqrt{2}}\right|^2 = 1/2$$

$$Pr(|10\rangle) = \left|\frac{1}{\sqrt{2}}\right|^2 = 1/2$$

$$Pr(|00\rangle) = Pr(|11\rangle) = 0$$

SCRIVERE IL GATE AGGREGATO



Si ripercorre il percorso al contrario:

$$U = HSX = \overbrace{\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}^{H} \overset{S}{\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}} \overset{X}{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}}$$

$$= \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} =$$

$$= \frac{1}{\sqrt{2}}\begin{pmatrix} i & 1 \\ -i & 1 \end{pmatrix}$$

PRODOTTO TENSORE

$$|1\rangle \otimes |0\rangle = |1\,0\rangle$$

Se voglio svolgere il prodotto tensore moltiplico gli elementi del primo operando con il secondo:

$$|1\rangle \otimes |0\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{bmatrix} 0 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ 1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

PROPRIETA':

- $|1\rangle \otimes -i\,|0\rangle = -i\,|1\rangle \otimes |0\rangle = -i\,|10\rangle$